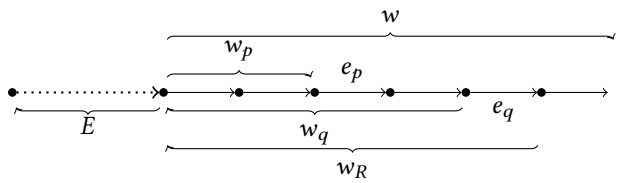

Fig. 6. Illustrating E, w, w_p, e_p, w_q, e_q and w_R.

Given these definitions, all the following are valid execution sequences:

$$\begin{aligned} E \vdash w &\longrightarrow E \vdash w_q.\widehat{e_q} \\ &\longrightarrow E.w_q \vdash \widehat{e_q} \end{aligned} \tag{8}$$

as well as the following:

$$\begin{aligned} E \vdash q \diamond w_q &\longrightarrow E.w_q \vdash q \\ &\longrightarrow E.w_q \vdash q.\widehat{e_q} && \text{by (7), (8) and Assumption 3.1} \\ &\longrightarrow E \vdash q.w_q.\widehat{e_q} && \text{by } E \vdash q \diamond w_q \\ &\longrightarrow E \vdash q.w_R && \text{by the definition of } w_R \end{aligned} \tag{9}$$

We also note that

$$p \notin WI_{[E]}(w_R) \text{ for all } p \in done(E) \qquad \text{by the construction of } w_R \tag{10}$$

As the next step in the proof of Claim 5.4, we establish the following claim:

Claim 5.5. *For all $p \in done(E)$ with $p \neq q$ we have $p \notin WI_{[E]}(q.w_R)$.*

Proof of Claim 5.5. We distinguish among the following cases:

1. If $e_p \neq e_q$, then w_p is a strict prefix of w_q. Therefore:

$$\begin{aligned} e_p \in w_q &\longrightarrow p \notin WI_{[E]}(w_q) && \text{by } E \nvdash p \diamond w_q \\ &\longrightarrow p \notin WI_{[E]}(w_q.q) && \text{by 5.1.2b} \\ &\longrightarrow p \notin WI_{[E]}(q.w_q) && \text{by } E.q.w_q \simeq E.w_q.q\text{, 5.1.1b} \\ &\longrightarrow p \notin WI_{[E]}(q.w_R) && \text{by 5.1.2b} \end{aligned}$$

2a. If $e_p = e_q$ and $E \nvdash p \diamond q$, then $p \notin WI_{[E]}(q.w_R)$ follows directly.
2b. If $e_p = e_q$, $E \vdash p \diamond q$, and $E \nvdash p \diamond q.w_q$, then $p \notin WI_{[E]}(q.w_R)$ follows directly.
2c. If $e_p = e_q$, $E \vdash p \diamond q$, and $E \vdash p \diamond q.w_q$, then:

$$\begin{aligned} E \vdash p \diamond q.w_q &\longrightarrow E \vdash p \diamond w_q.q && \text{by } E \vdash q \diamond w_q \\ &\longrightarrow E.w_q \vdash p \diamond q \end{aligned} \tag{11}$$

$$\begin{aligned} e_p = e_q &\longrightarrow E.w_q \nvdash p \diamond \widehat{e_q} \\ &\longrightarrow E.w_q \nvdash p \diamond q.\widehat{e_q} && \text{by (11) and Property (7) of Def. 3.2} \\ &\longrightarrow E \nvdash p \diamond w_q.q.\widehat{e_q} \\ &\longrightarrow E \nvdash p \diamond q.w_q.\widehat{e_q} && \text{by } E \vdash q \diamond w_q \\ &\longrightarrow p \notin WI_{[E]}(q.w_R) \end{aligned}$$